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9 July 1963

MILITARY SPECIFICATION

RATE INTEGRATING GYRO
MS 3391

This specification has been approved by
The Naval Air Systems Command Department of the Navy

1. SCOPE

1.1 Scope - This specification covers design requirements and all performance requirements for procurement of a rate integrating gyro.

2* APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-290	Nickel Plating (Electrodeposited)
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Corrosion Protection
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-T-5422	Testing, Environmental, Aircraft

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MIL-S-7742	Screw Threads; Standard Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-Q-9858	Quality Control-System Requirements
MIL-W-16878	Wire, Electrical, Insulation
MIL-M-45202	Magnesium Alloy, Anodic Treatment of
MIL-D-1000	Drawing, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of
MIL-STD-454	Soldering of Electrical Joints (REQ 5)
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS-33586	Metals, Definition of Dissimilar
MS-3391	Rate Integrating Gyro
NAS-3408	Box, Unit, Cushioned

2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specification required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Phila., Pa. 19120.

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3. REQUIREMENTS

3.1 Qualification - The gyroscope furnished under this specification shall be a product, samples of which have been tested, passed the Qualification Tests specified herein, and have been listed on or approved for listing on the applicable qualified products lists. (See 6.3)

3.2 Components - This gyroscope shall be of the rate integrating type. The major components of this gyroscope are gyro motor, gimbal, torquer, and pickoff.

3.3 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard (MS or AN) parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drwings.

3.3.2 Standard Parts - With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part number.

3.4 Materials - Materials shall conform to applicable specification and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Critical Materials - Non-critical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

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3.4.2 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the gyroscope may be subjected when in storage or during normal service life.

3.4.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS-33586 will not be used in intimate contact with each other.

3.4.3 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the gyroscope except where magnetic materials are required.

3.4.4 Fungus-Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.5 Corrosive Fumes - The materials as installed in the gyroscope and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.6 Finish - Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.4.7 Aluminum Parts - Where practical, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily, shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.4.8 Magnesium Parts - Where practical, magnesium alloy parts shall be plated in accordance with Specification MIL-M-3171A or MIL-M-45202 and of a class that is adequate to achieve the necessary degree of protection.

3.4.9 Steel Parts - Where practical, steel parts shall be plated in accordance with Specification QQ-N-290, QQ-C-320, or QQ-P-416, and of a class that is adequate to achieve the necessary degree of protection.

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3.4.10 Cable - The external wires in the cable shown on applicabl specification sheet shall conform to MIL-W-16878. The wire size shall be number 32 minimum. The leads shall be sheathed with a suitable jacket possessing sufficient flexibility to allow a bend of one inch radius without damage.

3.4.11 Test Fixture - A test fixture suitable for testing the gyro shall be used. The fixture shall be made of aluminum with a minimum outside dimension of 3 1/2 x 3 1/2 x 3 1/2 inches. Thermal insulation between the gyro and the test fixture shall be provided for all testing.

3.5 Design and Construction

3.5.1 General - The gyro shall be a single-degree-of-freedom instrument possessing the following basic components:

(a) An electrically-driven rotor, also referred to as spin motor.

(b) A gimbal, for support of the rotor, with freedom of rotation about an axis perpendicular to that of the supported rotor.

(e) A pickoff for development of a signal proportional to the angle of gimbal rotation about the axis of freedom.

(d) A torquer for development of torque applied to the gimbal about the axis of freedom.

In addition, other components are required to render the assemblage of the basic components operable. These other components may be variable in number and nature depending upon specific design approach adopted to achieve conformity to this specification.

The gyroscope shall be designed and constructed in accordance with the requirements of this specification and MS-3391. It shall be built to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation and service and a minimum of 2000 hours of service usage.

3.5.1.1 Storage - The storage life of the gyroscope shall be five years minimum when stored in the container supplied for shipment and exposed to temperature riot lower than -62°C nor higher than +95°C. When removed from storage, the gyro shall be capable of meeting all the performance requirements of this specification without maintenance.

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3.5.2 Functional Performance of Components - The following requirements shall apply to the components of the gyro.

3.5.2.1 Spin Motor Performance - The gyro spin motor shall operate from a 2-phase, 900 Hz supply with a time phase difference of 90° +20°. Each phase voltage shall be 6.5 ± 0.5 volts during normal operation but satisfactory operation shall be achieved over a range of 5.5 to 9.0 volts RMS. Operation of the motor shall be synchronous.

The impedance of each motor winding phase over all environmental conditions shall remain within the contour of Figure 5.

3.5.2.2 Gimbal Performance - The gyro gimbal shall possess sufficient rigidity to permit the gyro assembly to meet performance requirements. The assembly of spin motor, spin motor bearings and gimbal shall exhibit anise-elasticity not exceeding .00045°/sec/g^2. Gimbal damping and support shall be provided in the complete gyro by means of suitable fluid. Support by buoyant force in the fluid shall be such that gimbal pivot reactions at the axis of freedom shall not exceed 10% of the gimbal weight in air in a 1.0 ± 1% gravitational field (reckoned with respect to sea level, at the Naval Research Laboratories, Washington, D.C.).

3.5.2.2.1 Angular Acceleration Sensitivity - The equivalent rate signal developed by the gyro in response to angular accelerations about the output axis shall not exceed .003 rad/sec// rad/sec^2.

3.5.2.3 Torquer Performance - The torquer shall be a d'Arsonval type. The permanent magnet which forms a part of this component shall be stabilized against stray fields of 12.5 gauss maximum to the extent that exposure of the gyro assembly to fields of this magnitude shall not result in permanent departure of the torquer scale factor from the specified accuracy limits.

The torquer shall be connected in series with an external resistor as shown in Figure 4. Value of the external resistor shall be determined by the desired Rate Loop Scale Factor (see 3.5.3.3) and shall be in accordance with the following table. The series resistance of the torquer and the external resistor shall not be less than the values listed in the following table.

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Rate Loop Scale Factor* (VDC/DEG/SEC)	External Series Resistor (OHMS ±0.1%)	Minimum Torquer Circuit Resistance (OHMS)
1.0	1350	1600
0.5	420 or 500**	800

*Rate loop scale factor is defined as $\frac{H}{\frac{180}{\pi} K_T}$, where:

H = Angular momentum of gyro wheel (dyne cm/rad/sec).

K_T = Torquer scale factor (dyne cm/amp)/(torquer resistance + external series resistor-ohms).

** The use of a 420 Ω series resistor requires a jumper between pins J and T of the gyro connector as noted on electrical schematic, Figure 3.

3.5.2.4 Pickoff Performance - The pickoff shall be an AC moving coil device designed to operate on 900 Hz, 29.5 +0.5 volts RMS, single-phase primary excitation. The pickoff secondary loading is 2700 ohms resistive for the performance characteristics described in Paragraph 3.5.3. The pickoff primary impedance over all environmental conditions shall be within the limits of Figure 6.

The pickoff input power shall not exceed 1.5 watts. The pickoff secondary resistance shall lie within the limits of 289 ohms to 450 ohms at room temperature for a heaterless gyro, or at operating temperature for a heated gyro. The inductance shall not exceed 10 mh. Total quadrature and noise with the pickoff at null shall not exceed 15 mv RMS. The characteristics of the pickoff shall not be permanently affected by excitation voltages within the range of zero to 37 volts RMS.

3.5.2.5 Heater - When required by the applicable Military Standard (MS) the heater shall be non-inductively wound. Its DC resistance value shall be not less than 50 ohms when measured at 22 +5°C. The heater shall operate satisfactorily from a power source-with a peak output of 115 volts RMS at 400 Hz and a maximum power output of 175 watts. The heater shall draw maximum power for no more than 5 minutes and the continuous power needed by the heater shall be less than 100 watts. The heater shall be capable of warming the gyro up such that it meets the warm-up time requirements of this specification, in addition to maintaining the gyro at its operating temperature.

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3.5.2.6 Temperature Regulator - If a heater is used on the gyro, control of the external heater power source shall be provided by a temperature sensor non-inductively wound within the envelope of the gyro as defined in the specification sheet. The sensor element shall be wound on the same form which holds the gyro heater and shall have intimate thermal contact with the heater. The temperature sensor shall have a resistance of 780 +2 ohms at the gyro operating temperature with a positive temperature coefficient and a sensor scale factor of 1.5 +10%, -20% ohms per degree Fahrenheit. The external power source will supply 16 to 22 watts per ohm of sensor resistance change into a 57 ohm heater load from a maximum of 175 watts to a minimum of less than 2 watts as shown in Figure 7.

3.5.2.7 Heater Protection - If a heater is used in the gyro, a protective circuit shall be provided which will interrupt the power to the heater circuit before the temperature of the gyro reaches a value high enough to cause damage to the gyro. The protective circuit shall be contained within the envelope of the gyro as defined in the specification sheet.

3.5.3 Performance Characteristics - With the gyro connected as shown in Figure 3 and operating in the closed loop mode (unless otherwise specified) by means of a suitable restraint amplifier, the following characteristics apply.

3.5.3.1 Rate Loop Scale Factor - the gyro, when operated closed loop, shall exhibit a rate loop scale factor of 1 vdc/deg/sec or 0.5 vdc/deg/sec when used with series resistors listed in 3.5.2.3 over a range of zero to at least ± 90 deg/sec. When the qyro is tested in accordance with 4.8.1.3, the output rate signal shall be within +1% of the above scale factors.

3.5.3.2 Linearity - The gyro linearity shall be as follows:

Angular Rate (Deg/Sec)	Linearity Error (Max)
0 - 5	0.02°/sec or 1% of true rate whichever is greater
5 - 60	1% of true rate
60 - 90	2% of true rate

3.5.3.3 Total Gyro Drift Rate - Test in accordance with 4.8.1.7.

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3.5. 3.3.1 Zero Rate Signal (ZRS) - The gyro output signal for a zero input rate shall not exceed the value corresponding to 0.020 deg/sec in each of the six gyro positions corrected for–earth's rate.

3.5.3.3.2 Zero Rate Uncertainty (ZRU) - The ZRS shall be stable and repeatable in any given gyro position within a band of 0.010 deg/sec within any one test sequence.

3.5.3.3.3 Zero Rate Stability - The ZRS drift shall not exceed 0.010 deg/sec.

3.5.3.4 Cross-Coupling - The maximum cross-coupling shall not exceed ±0.5% of the input rate when mounted with either orientationpin mouning flange or orientation notch mounting flange combination. Test in accordance with 4.8.1.5.

3.5.3.5 Hysteresis - After being subjected to alternate clockwise and counterclockwise rates, parallel to the input axis, of 100 deg/sec to 150 deg/see, the width of the hysteresis measurement shall not exceed the equivalent of 0.020 deg/sec. Test in accordance with 4.8.1.8.

3.5.3.6 Warm-up Time - The gyro shall meet all functional requirements of this specification within 5 minutes of the time power is applied.

3.5.3.7 Dynamic Response Parameters - The following gyro parameters shall be held to the tolerances given when operating over the ambient temperature range specified in MIL-E-5400, Class II equipment.

3.5.3.7.1 Gyro Scale Factor - The gyro scale factor is defined as the ratio of the gyro output angle, as measured by the pickoff output, to gyro input angular displacement when operating in an open loop mode. the GSF is expressed as:

$$\frac{E_{out}}{\theta_{in}} = \frac{H}{B} \times K_p$$

where: H = angular momentum (dyne cm-sec)

K_P = pickoff scale factor (MVrms/mrad)

B = coefficient of damping (dyne cm-sec)

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E_{out} = gyro pickoff output (MVrms)

e_{in} = gyro input angular displacement (deg).

When tested in accordance with 4.8.1.6.1, the gyro scale factor shall lie between the limits of 0.20 to 0.36 volts RMS/deg input.

3.5.3.7.2 Gimbal Corner Frequency (B/J) - the gimbal comer frequency is defined as the ratio of the damping coefficient to the effective gyro gimbal moment of inertia:

where: B = coefficient of damping (dyne cxn-sec)

J = Effective gimbal moment of inertia (dyne $cm\text{-}sec^2$).

When tested in accordance with 4.8.1.6.2, the gimbal corner frequency shall lie between the limits of 245 to 495 rad/sec. With a gyro scale factor below 0.22, the gimbal corner frequency shall not be less than 300 rad/sec.

3.5.3.8 Run-up Time - The gyro motor run-up time shall be measured to determine that the motor has sufficient torque margin. When tested in accordance with 4.8.1.9.1, this time shall not exceed 30 seconds.

3.5.4 Reliability - The gyro is required to have a 90% probability of satisfactorily operating for 2000 hours. Satisfactory operation consists of meeting all the requirements of paragraphs 4.8.1 and 4.8.2.

3.5.5 Auxiliary Electronic Equipment

3.5.5.1 Restraint Amplifier - A suitable restraint amplifier shall be used for closed loop mode operation.

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3.5.5.2 Gyro Inputs

Pin	Applied Power	Circuit Application
A	-	Pickoff Output
B	-	Pickoff Output, Gnd. Ref.
C	29.5 ±0.5V RMS, 900 ±0.1% Hz	Pickoff Primary
D	6.5 ±0.5V RMS, 900 ±0.1% Hz	Spin Motor Excitation,
E	-	Common for Motor, Pickoff and Sensor (Case Optional)
F	6.5 ±0.5V RMS, 900 ±0.1% Hz	Spin Motor Excitation, 90° Leading Phase
H	-	Sensor (optical)
Y	-	Torquer
K	-	Torquer, Gnd. ref.
L,M	0-115V RMS 400 +5% Hz	Heater (optional)
N	-	Shield Ground
T	-	Torquer (Pin T connected to Pin J, Optional)

3.5.5.3 Output Loads and Filter - When tested, the gyro output shall be connected to simulated loads as shown in Figure 4.

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3.5.6 Additional Electrical Requirements

3.5.6.1 Positive input rate shall result in CCW gimbal rotation as viewed from cable end of gyro. Positive spin motor rotation (see Figure 1) shall result when excited so that the voltage at terminal E' leads terminal D by 90° ± 20°, terminal E being common to both phases. With positive spin motor rotation, a CCW input rotation shall cause the output voltage on the pick-off secondary (terminal A) to be in Phase with and lead the primary (terminal C) by 25° ±5°, terminal E and B being common. When no input rate is present and the torquer is excited so that terminal K is positive with respect to terminal J, the gimbal shall rotate such that the pickoff output voltage is in phase with and leading the input by 25° ±5°. See 4.8.1.2.

3.5.6.2 Primary Power Transients - The gyro shall not be damaged when subjected to power source transients of MIL-STD-704.

3.5.6.3 Dielectric Strength - When tested in accordance with 4.8.3, the electrical insulation shall not break down at the voltages specified.

3.5.6.4 Insulation Resistance - when tested in accordance with 4.8.4, the insulation resistance shall not be less than 50 megohms for any test specified.

3.5.6.5 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.5.6.6 Electronic Parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on the approved lists shall not be used unless approved by the procuring activity.

3.5.7 Additional Mechanical Requirements

3.5.7.1 Dimensions - The dimensions of the gyro shall be in accordance with the applicable Military Standard.

3.5.7.2 Soldering - Soldering shall be accomplished in accordance with Specification MIL-S-6872 and MIL-STD-454.

3.5.7.3 Weight - The gyro mass, excluding cable and connector shall not exceed 210 grams. The gyro weight with cable and connector shall not exceed 284 grams.

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3.5.7.4 Screw Threads - Unless otherwise specified, the threads of all machine screws 0.060 inches or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.6 Environmental - The gyro shall operate as specified when subjected to any of the following environmental conditions. There shall be no electrical or mechanical damage as a result of the specified tests. There shall be no discontinuity of electrical connections during or after the tests. Unless otherwise specified, acceptability limits shall consist of conformance to the requirements of the following paragraphs:

3.5.3.2	Linearity
3.5.3.3	Total Gyro Drift Rate
3.5.3.5	Hysteresis
3.5.3.7	Dynamic Response
3.5.3.8	Run-up Time

3.6.1 Low Temperature Operation - Test in accordance With 4.8.2.1

3.6.2 High Temperature Operation - Test in accordance with parts 1 and 2 of 4.8.2.2.

3.6.3 Vibration - The part shall be tested in accordance with the vibration sampling and vibration endurance tests of 4.8.2.3.

3.6.4 Shock - Test in accordance with 4.8.2.4.

3.6.5 Radio Noise - The gyro shall not exceed the generated interference requirements of MIL-STD-461, for Class IIB equipment. Test in accordance with 4.8.2.5.

3.6.6 Temperature and Altitude - Test in accordance with 4.8.2.6. During test satisfactory performance shall be determined by conformance with the drift requirements of 3.5.3.3 as tested by 4.8.1.7.3.

3.6.7 Salt Spray - Test in accordance with 4.8.2.7. After salt spray test, test the gyro to determine conformance with 4.8.4 (a), (d), and (h), and 3.5.3.3 as shown in 4.8.1.7.1.

3.6.8 Multiple Cold Soak - Test in accordance with 4.8.2.8. After each cold soak, determine that the gyro conforms to the requirements of 3.5.3.3.

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3.6.9 Anisoelasticity - Test in accordance with 4.8.2.9. The difference in the output rate signals between vibration applied and not applied shall not exceed a rate output equivalent to ± 0.045 deg/sec or a coefficient of ± 0.00045 deg/sec/g^2 RMS .

3.6.10 Warm-up Time - When the gyro is tested in accordance with 4.8.2.10, the warm-up time shall not exceed the time specified in 3.5.3.6.

3.6.11. Life - The gyro shall be tested in accordance with 4.8.2.11 for a total on time of 2000 hours. In addition to the acceptance criteria of 3.6, the part shall be tested to show conformance to the output sensitivity requirements of 3.5.3.1. During life test the part shall be tested at intervals as shown in 4.8.2.11.

3.7 Identification of Product - Gyros shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship - The gyroscopes, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.9 Cleaning - The gyro shall be thoroughly cleaned and loose spattered or excess solder, flux, metal chips and other foreign material shall be removed,

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Responsibility - Unless otherwise specified quality control system requirements are per MIL-Q-9858. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Tests - Gyros covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Qualification Tests (4.2.1)

(2) Initial Production Tests (4.2.2)

(3) Acceptance Tests (4.2.3)

(4) Life Tests (4.8.2.11)

4.2.1 Qualification Tests for QPL - Twelve gyros manufactured in accordance with the requirements of this specification, submitted at no cost to the Government, shall be subjected to the tests described in Schedule 1 at an activity designated by the agency responsible for granting qualification approval. All tests must be satisfactorily complete prior to qualification approval being granted (see 6.3). Production of gyros prior to approval shall be at the manufacturer’s risk. Gyros submitted to this test shall not be shipped as production parts. See 5.4 for qualification part marking requirements

4.2.2 Initial Production Tests - Three of the first ten production gyros shall be selected, and sent at the contractor's expense to a designated Government laboratory, and subjected to tests in accordance with the outline in schedule 2. The gyros shall be selected by the procuring activity after the gyros have successfully passed all inddividual tests. No other tests shall be conducted on the gyros prior to starting the Initial Production Tests. The qualification samples shall not be selected for this test.

4.2.3 Acceptance Tests - The contractor shall furnish samples to be subjected to the following tests:

(1) Individual Tests (4.2.4)

(2) Sampling Tests (4.3)

4.2.4 Individual Tests - Each gyro, to be delivered under the contract, shall be subjected to the functional tests per paragraph 4.8.1. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability.

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4.2.5 Quality Assurance Data - All failures experienced on delivered gyros shall be recorded and submitted in accordance with paragraph 6.5. For defective parts action paragraph 4.5 sections (1) and (3) apply.

4.3 Sampling Tests - After initial production test (4.2.2) one gyro from each succeeding lot (4.6) shall be selected at random and subjected to the tests specified in Schedule 3, at an activity designated by the procuring activity. see 5.6.

4.4 Design Changes - At any time when a major design change is incorporated, Initial Production Tests (4.2.2) shall be conducted.

4.5 Defective Parts - When a defective gyro is found during the performance of tests described in paragraphs 4.2.2, 4.2.5; or 4.3, the manufacturer shall take the following action:

(1) Determine the cause of failure

(2) Take the necessary actions to correct or eliminate the defect from all gyros on hand and, if so directed by the procuring activity, from those gyros previously shipped, The cost of this entire effort to be paid for by manufacturer.

(3) Submit to the procuring activity for approval the proposed corrective actions to prevent recurrence of the defect.

4.6 Production Lot Size - The lot size for the purpose of sampling tests shall be 100 gyros or any fraction thereof remaining on the contract.

4.7 Failure. - A gyro failure is any malfunction which causes the gyro to fail to meet the minimum performance requirements specified herein.

4.8 Test Methods -

4,8.1 Functional Tests - The individual tests of the gyro shall consist of the following tests under the test conditions of 4.9.

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4. 8.1.1 Inspection - The gyro shall be inspected to determine compliance with this specification with regard to material, workmanship and marking.

4.8.1.2 Phasing - A two part test will be performed to insure that the gyro loop and the spin motor are properly phased.

4.8.1.2.1 Gyro Loop - Place the gyro on a flat horizontal surface such as a single axis rate table. Position the gyro so that the input axis is vertical and the orientation pin as viewed from the cable end of the gyro is in the 3 o'clock position and parallel to the plane of the table. With the gyro operating, stimulate the gyro with a brief CW or CCW motion. The gyro output shall stabilize near zero as opposed to a saturated condition.

4,8.1.2.2 Spin Motor Phasing - Open the gyro loop by disconnecting pin J and pin T with the gyro placed on the rate table with the orientation pin, as viewed from the cable end of the gyro, in the 12 o'clock position, and with the motor excitation voltage on pin F leading the motor excitation voltage on pin D, rotate the rate table in a CW direction. 'The pickoff output shall stabilize in a nearly nulled condition as opposed to a saturated output condition.

4.8.1.3 Output Sensitivity - The gyro shall be mounted on a single axis rate table. As viewed from the cable end of the gyro, the orientation pin shall be at the 3 o'clock position and parallel to the plane Of the table (spin axis longitudinal, input axis vertical). The 3 o'clock orientation shall be established by means of a suitable gyro fixture within 5 minutes of arc. Rotate the table at a rate of 20 degrees per second ±0.1% both CW and CCW. The output rate signal shall be in accordance with 3.5.3.1.

4.8.1.4 Linearity of Rate Loop Scale Factor - Use a rate table and gyro orientation as in 4.8.1.2 for the following applied angular rates both CW and CCW. Measure and record the output rate signals which shall be in accordance with 3.5.3.2. Use applicable external series resistor as specified in 3.5.2.3 for 1.0 volt/deg/sec rate loop scale factor and 0.5 volt/deg/sec rate loop scale factor.

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Input Rate (deg/see)	Output Rate Signal (VDC) (Ref)	
	1.0 VDC/deg/sec	0.5 VDC/deg/sec
5 ± .01	5.00 ± 05	2.50 ± .025
20 ± .02	20.00 ± .20	10.00 ± .10
40 ± .04	40.00 ± .40	20.00 ± .20
60 ± .06	60.00 ± .60	30.00 ± .30
90 ± .09	90.00	45.00 ± .90

4.8.1.5 Cross-Coupling

4.8.1.5.1 Notch Alignment - Align the gyro on the rate table, indexing on the gyro notch so that the table axis and the gyro notch and flange centers are aligned.

(a) Mark the position of the fixture on the rate table. Rotate at 90 ± 5 deg/sec CW. Record the output reading.

(b) Rotate the gyro fixture 180 degrees about the spin axis (do not invert) and align with the marks on the table. Rotate at 90 ±5 deg/sec CW. Record the output reading.

(c) Repeat (a) in the CCW direction. Record.

(d) Repeat (b) in the CCW direction. Record,

(e) Call the algebraic average of (a) and (b) readings A. Call the algebraic average of (c) and (d) readings B. Compute the algebraic difference $\frac{A-B}{2}$, which shall be the true cross coupling. Record. The true cross coupling shall not exceed the limit of 3.5.3.4.

4.8.1.5.2 Pin Alignment - Align the gyro on the rate table indexing on the orientation pin so that the table axis and the pin and the gyro flange centers are aligned. Repeat procedures (a), (b), (c), (d), and (e) above.

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4. 8.1.6 Dynamic Response Parameters

4.8.1.6.1 Gyro Scale Factor - Orient the gyro as in 4.8.1.2.1. Rotate the gyro through 0.5 ± 0.01 angular degrees with pin J and pin T of the gyro disconnected and record the pickoff signal. This test shall be repeated three (3) times with a CW input rotation and three (3) times with a CCW input rotation, The pickoff output value must be recorded within three (3) seconds after applying the full input axis motion. Care must be exercised to perform the test rapidly enough to prevent drift away from null after disconnecting pin J and pin T prior to applying the input axis motion. In accordance with 3.5.3.7.1 the average of the six readings shall be 0.100 to 0.180 volt RMS.

4.8.1.6.2 Gimbal Corner Frequency Test - The gimbal corner frequency shall be measured with the gyro connected in a circuit in accordance with Figure 8. The gimbal corner frequency in radians/second is the frequency at which the phase lag between the torquer drive current and the envelope of the pickoff output. voltage is 135 degrees.

Momentarily close the normal gyro loop to null the gyro gimbal position. De-energize the spin motor. Apply 15.O ± 1.O volts RMS, 10,000 Hz to the pickoff primary. Energize the torquer with a fixed frequency sinusoidal voltage source between 50 and 60 Hz. Increase the amplitude of the drive voltage until the torquer current is 60 ± 1 ma. Adjust the phase shifter until its output is in phase with the envelope of the pickoff output as indicated on the oscilloscope. (A slight adjustment of the 10,000 Hz frequency may be necessary to make the scope display stationary.) Algebraically add the phase angle voltmeter reading to 9.0° and divide the tangent of the resultant into the product of 6.28 times the fixed excitation frequency of the torquer. The devidend equals the corner frequency in radians/second and shall conform to the requirements of 3.5.3.7.2.

4.8.1.7 Total Gyro Drift Rate - The following tests shall be conducted to determine compliance with 3.5.3.3.

4.8.1.7.1 Zero Rate Signal (ZRS) - Energize the gyro and begin testing within five minutes. Record three readings in each of the six cardinal gyro positions. Reading shall be taken within 20 seconds at each position. Refer to Figure 2. The ZRS shall not exceed ±0.020 deg/sec when corrected for earth’s rate.

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4.8.1.7.2 Zero Rate Uncertainty - When testing in accordance with paragraph 4.8.1.7.1, the difference of the three ZRS readings in the same position shall not exceed 0.01 deg/sec within any given test.

4.8.1.7.3 Zero Rate Stability - with the gyro in Position 1, the ZRS drift shall be monitored for ten minutes (3 minutes for Acceptance Tests). The allowable drift shall not exceed 0.01 deg/sec. The total ZRS shall not exceed the value specified in 3.5.3.3.

4.8.1.8 Hysteresis - with the gyro energized, align the gyro by means of mounting fixture so that the input axis is parallel to the axis of rotation of the turn table.

(a) With the gyro in Position 1 of Figure 2 free-spin the turn table alternately CW and CCW with a minimum of 100 deg/sec and a maximum of 150 deg/sec. Allow the table to coast to a smooth stop after each spin and record the total drift rate within 20 seconds. A total of six readings (3 CW and 3 CCW) shall be taken for the average referred to in (c) below.

(b) Rotate the gyro 180 degrees about its output axis and repeat (a).

(c) In accordance with 3.5.3.5, the readings of (a) and (b) shall not exceed that output value corresponding to 0.02 deg/sec when corrected for earth's rate and the difference between the average CW and the average CCW shall not exceed that output value corresponding to 0.02 deg/sec.

4.8.1.9 Spin Motor Run-up

4.8.1.9.1 Spin Motor Run-up Time Measurement - Spin motor run-up time shall be defined within the context of the following measurement description. Energize the motor with 6.5 +0.5 volt RMS, 900 Hz ± 0.1% per phase. Check the time required For the spin motor to achieve synchronous speed three times from a stalled condition. Record. The value shall be in accordance with 3.5.3.8.

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4.8.1.9.2 Spin Motor Torque Margin - Energize the motor with 5.1 ±0.1 volt RMS, 900 Hz per phase as described in paragraph 4.9. The spin motor shall obtain synchronous speed per the following formula:

$$T = (K)(H)$$

T = torque margin (see)
H = angular momentum (dyne cm-see)
$K = 2 \times 10^{-3}$ (dyne cm)$^{-1}$

Repeat three times at two minute intervals and record. (This is a vendor preshipment requirement only).

4.8.2 Environmental Tests - Unless otherwise specified, after each of the environmental tests the gyro shall be tested in accordance with the following and shall meet the requirements indicated in the test paragraphs.

4.8.1.4	Linearity
4.8.1.6.1	Gyro scale factor
4.8.1.6.2	Gimbal corner frequency
4.8.1.7	Total gyro drift rate
4.8.1.8	Hysteresis
4.8.1.9.1	Run-up time

4.8.2.1 Low Temperature Operation - The gyro shall be mounted in a suitable test fixture and placed in a test chamber in Position 1. With the gyro non-operating, adjust the test chamber temperature to -62°C ±2°C. The gyro temperature shall be stabilized and maintained for at least two hours. With the gyro non-operating, adjust the chamber temperature to -54°C ±2°C. Monitor the pickoff resistance untill the gyro attains -54°C ±2°C. to Energize the gyro. Perform the following tests after the gyro has been energized for five minutes: 4.8.1.4, 4.8.1.6, 4.8.1.7, 4.8.1.8, and 4.8.1.9.1. Should the time required to check the equipment exceed 15 minutes beyond the warm-up time, the equipment shall again be stabilized at -54°C ±2°C and the operational check continued.

4.8.2.2 High Temperature Operation

4.8.2.2.1 High Temperature Operation Part I - The gyro shall be mounted in a suitable test fixture and placed in a test chamber in Position 1. With the gyro non-operating,

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adjust the chamber temperature to 71°C ±2°C. After the gyro temperature has stabilized, energize the gyro for five minutes and perform the following tests: 4.8.1.4, 4.8.1.6, 4.8.1.7.3, and 4.8.1.9.1. The gyro shall continue to operate for a total of four hours. Prior to the end of the four hours, repeat the above tests.

4.8.2.2.2 High Temperature Operation Part II - The gyro shall be mounted in a suitable test fixture and placed in a test chamber in Position 1. With the gyro non-operating, adjust the chamber temperature to 95°C ±2°C. After the gyro temperature has stabilized, energize the gyro for five minutes and perform the following tests: 4.8.1.4, 4.8.1.6, 4.8.1.7.3, and 4.8.1.9.1. The gyro shall continue to operate for 30 minutes total. De-energize the gyro for 15 minutes. This procedure shall be performed to complete four cycles.

4.8.2.3 Vibration

4.8.2.3.1 Vibration Sampling - This vibration test shall consist of a frequency survey with vibration applied for 10 minutes in each of the three mutually perpendicular principal axes. (Refer to Figure 1.) The gyro shall be subjected to a vibration of 0.1 inches double amplitude from 5 to 14 Hz, 1g from 14 to 23 Hz, 0.036 inches double amplitude from 23 to 52 Hz, 5g's from 52 to 2000 Hz.

4.8.2.3.2 Vibration Endurance

(a) Vibrate in each of the three mutually perpendicular axes for six hours per axis over the frequency range of 10 to 2000 Hz. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (10 - 2000 - 10) will consume approximately 30 minutes. The level of vibration shall be as follows:

10	-	30Hz at 0.3" DA
30	-	50Hz at ± 14g's
50	-	125Hz at ± 20g's
125	-	200Hz at ± 14g's
200	-	500Hz at ± 10g's
500	-	2000Hz at ± 5g's

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The gyro shall be non-operating for random 15 minute periods and the total non-operating time shall be 25% of the total test time. During the operating period, the zero rate signal output shall be monitored for intermittent operation.

(b) Vibrate for five minutes along each of the three principal axis at 30g's over the frequency range of 50-125 Hz with the gyro operating.

4.8.2.4 Shock - The gyro shall be securely fastened using the normal mounting means. The gyro shall be fully energized. Test in accordance with Section 4.3 of MIL-T-5422.

4.8.2.5 Radio Noise - The generation of radio interference by the gyro shall be tested as required by MIL-STD-461, Table I Category IIB, and shall not exceed the limits specified in Section 6 of MIL-STD-461.

4.8.2.6 Temperature and Altitude - Test in accordance with Section 4.1 of MIL-T-5422, Class 2 equipment.

4.8.2.7 Salt Spray - Test in accordance with Section 4.5 of MIL-T-5422. After salt spray, test in accordance with paragraph 4.8.4, Sections (a), (d), and (h) and 4.8.1.7.1.

4.8.2.8 Multiple Cold Soak

(a) The gyro shall be mounted in a suitable test fixture and placed in a chamber that has been stabilized at -62°C ±2°C.

(b) After the gyro temperature has stabilized at -62°C ±2°C for two hours, remove the gyro from the test chamber.

(c) With the gyro in the ambient room temperature allow it to warm up to -54° ±2°C. At this point, energize the gyro.

(d) After five minutes of energizing, measure the zero rate according to paragraph 4.8.1.7.

(e) Repeat steps (a), (b), (c) and (d) five times. Total six times.)

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4.8.2.9 Anisoelasticity - Position the gyro in a rigid holding fixture and position it with the ooutput axis horizontal and the input axis 45° ±5° from vertical. Energize the gyro and hold in a closed loop. With the axis of vibration in the vertical direction, perform a vibration scan from 200 to 500 Hz at a 2g RMS level to determine any resonant points. At a point at least 50 Hz from any resonant point, vibrate the gyro at 10g RMS ±0% and record the gyro output rate signal. Reduce the vibration level to zero and record the output rate signal. The difference in output rate signals shall not. exdeed the. limits of 3.6.9.

4.8.2.10 Warm-up Time - The procedure for the warm-up is as follows starting with the gyro stabilized at -54°C ±2°C:

(a) Immediately position the gyro on a rate table as in 4.8.1.3.

(b) Connect the gyro into the closed loop but with no energization.

(c) Start timing upon energizing the heater circuit. If no heater is used, apply power to all circuits after 40 seconds.

(d) Upon reaching 40 ± 10 seconds of heater energization, energize all remaining gyro circuits and simultaneously apply a 20° +0.1% per second table rate.

(e) Stop timing when the gyro output rate signal falls within 1% of the value obtained in 4.8.1.3. The total elapsed time is the warm-up time and shall conform to the requirement of 3.6.10.

4.8.2.11 Life Test. - During life test, the spin motor shall be excited with 6.5 ±0.5 volt RMS per phase. The gyro may be operated in a closed rate loop. The gyro shall be operated for a period of four hours followed by an off time of 10 minutes or the time necessary for the gyro temperature to decrease to a value which is 10°C above ambient temperature, whichever is greater. Life test shall consist of repeating the above sequence for a total running time of 2000 hours as required by 3.6.11. During test, the gyro performance shall be verified by the tests shown in 4.8.2 and 4.8.1.3. For the first 800 hours of test, the performance shall be verified by test every 100 hours. During the remainder of life test, performance shall be verified every 200 hours.

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4.8.3 Dielectric Strength - The gyro shall be tested by application of the voltages specified below. The voltage shall be approximately 60 Hz applied for 10 seconds between isolated windings and between windings and case. The part shall meet the requirements of 3.5.6.3.

(a) Heater to ground	500 volts RMS
(b) Sensor to ground	500 volts RMS (prior to final cabling)
(c) Pickoff Primary to ground	250 volts RMS (at subassembly)
(d) Pickoff Secondary to ground	100 volts RMs
(e) Torquer to ground	500 volts RMs (at subassembly) 250 volts RMS (at final assembly)
(f) Spin Motor to ground	100 volts RMS (at subassembly)
(g) Associated Circuitry	250 volts RMs (at subassembly]
(h) Between any two isolated windings	100 volts RMs

4.8.4 Insulation Resistance - The gyro shall be tested by application of the DC voltages specified below. The insulation resistance shall be measured between isolated windings and between windings and case as shown with voltage applied for one minute. Insulation Resistance shall be in accordance with 3.5.6.4.

(a) Heater to ground	250 volts DC
(b) Sensor to ground	250 volts Dc (at subassembly)
(c) Pickoff primary to ground	250 volts DC (at subassembly)

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(d)	Pickoff secondary to ground	100 volts DC
(e)	Torquer winding to ground	250 volts DC
(f)	Spin motor to ground	100 volts DC (at subassembly)
(g)	Associated Circuitry	250 volts DC (at subassembly)
(h)	Between any two isolated windings	100 volts Dc

4.9 Test Conditions - Unless otherwise specified, the following test conditions shall prevail:

4.9.1 Temperature - The ambient temperature shall be 25°C +5°C.

4.9.2 Atmospheric Pressure - The atmospheric pressure shall be a nominal 30" hg.

4.9.3 Relative Humidity - The relative humidity shall be the normal room ambient (50% nominal).

4.9.4 Gyro Spin Motor Excitation - 6.5 ±0.5 volt RMS unless otherwise specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The gyroscope shall be packaged in accordance with Specification MIL-P-7936, Level A or B as specified in the contract. For Level A packaging, the method of preservation shall be in accordance with Specification MIL-P-116 Method 1A-5 without preservative compound, using re-usable containers.

5.2 Packing - The gyroscope shall be packed in accordance with Speification MIL-P-7936, Level A, B, or NAS 3408 (or equivalent) as specified in the contract.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

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5.4 Marking of gyros submitted for QPL approval - Each gyro shall be plainly identified by securely attached durable tags marked with the following information:

"SAMPLE FOR QUALIFICATION TESTING

Gyroscope, Rate Integrating (type number, if any) Submitted. by (Manufacturer's name, date) for qualification tests.

In accordance with MIL- under authorizing (reference the letter authorizing the tests)."

5.5 Marking of gyros submitted for Initial production testing - Each gyro shall be plainly identified by securely attached tags marked with the following information:

"SAMPLE FOR INITIAL PRODUCTION TESTING

Gyroscope, Rate Integrating (type number and serial number)

Submitted by (Manufacturer's name, date) for Pre-production Testing in accordance with MIL- under contract number _____.

5.6 Marking of gyros submitted for Production Testing Samples submitted shall be plainly identified by a securely attached durable tag marked with the following information:

"SAMPLE FOR PRODUCTION TESTING

Gyroscope, Rate Integrating (Military part number serial number)

Submitted by (Manufacturer's name, date) for production testing in accordance with MIL- under contract number _____."

Upon successful completion of the sample test, the identification tags shall be removed and the gyro shipped to the point of destination as designated by the procuring activity.

6. NOTES

6.1 Intended Use - The rate integrating gyro covered by this specification will produce a signal proportional to angular velocity about a single axis in a rate loop configuration.

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6.2 Ordering Data - Procurement documents should specify the following:

(a) Military Part Number

(b) Title, number, and date of this specificaiton.

(c) Name of the laboratory where samples shall be submitted.

(d) Levels of packaging and packing desired.

6.3 Qualification - With respect to products requiring qualification awards will be made only for such products as have, prior to time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Washington, D. C., and information pertaining to qualification of products may be obtained from that activity.

6.4 Provisions for Initial Production Tests - When a contractor is in continuous production of these gyroscopes from contract to contract, submission of further initial production samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of initial production samples or the waiving of initial production tests does not preclude the requirements of acceptance testing.

6.5 Failure Data - All delivered gyros that fail during the first 2000 hours or 3 years of operation, will be reported to the procuring activity on a monthly basis.

6.6 Precedence of Documents - When the requirements of the contract, this specifcation, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

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(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specification shall apply to the extent specified.

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6.7 Rate Integrating Gyro Identification (Information Only)

Manufacturer & Designations	Alignment Method	Scale Factor	Heater	Application
LSI 1903A, T1903A, 139987-01-01 (FSN 1430-072-4522-M558)	Pin	1 volt/deg/sec	No	APQ-72, APQ-100 APQ-109
LSI 1903B, 141450-01-01 Note (1), (2)	Pin	1 volt/deg/sec 0.5 volt/deg/sec	No	APQ-72, APQ-100 APQ-109, APQ-120 APG-59, Prototype
*LSI 1903H, 146600-01-01 Note (3)(4)	Pin and Notch	1 volt/deg/sec 0.5 volt/deg/sec	No	APQ-72, APQ-100 APQ-109, APQ-120 APG-59
Honeywell GG250A2 (FSN 6615-871-8648-M558)	Pin	1 volt/deg/sec	Yes	APQ-72, APQ-100 APQ-109
Honeywell GG250A4 Note (1), (2)	Pin	0.5 volt/deg/sec	Yes	APG-59 Prototype
Honeywell GG250A6 (FSN 1430-921-5729-M558)	Pin	1 volt/deg/sec 0.5 volt/deg/sec	Yes	APQ-72, APQ-100 APQ-109, APQ-120 APG-59
Honeywell GG250A11 (FSN RQ1430-933-3639-BFEE)	Pin and Notch	1 volt/deg/sec 0.5 volt/deg/sec	Yes	APQ-72, APQ-100 APQ-109, APQ-120 APG-59
*Honeywell GG250A12 Note (3)(4)	Pin and Notch	1 volt/deg/sec 0.5 volt/deg/sec	Yes	APQ-72, APQ-100 APQ-109, APQ-120 APG-59

*These items described by this specification. (See Current QPL.)

NOTE: (1) Contractor Furnished Equipment; (2) Prototype; (3) FSN to be assigned
(4) See "Military Spec Sheet" for Military Part Number

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TEST DESCRIPTION	REFERENCE PARAGRAPH	PREPRODUCTION SAMPLE												
		1	2	3	4	5	6	7	8	9	10	11	12	
Functional Test	4.8.1.2, 4.8.1.3, 4.8.1.4, 4.8.1.5, 4.8.1.6, 4.8.1.7, 4.8.1.8, 4.8.1.9.1	X	X	X	X	X	X	X	X	X	X	X	X	
Low temperature operation	4.8.2.1							X	X					
High temperature operation	4.8.2.2							X	X					
Radio noise	4.8.2.5										X			
Temperature and altitude	4.8.2.6					X	X							
Salt spray	4.8.2.7									X				
Vibration endurance	4.8.2.3.2							X	X					
Life test	4.8.2.11	X	X	X	X									
Multiple cold soak	4.8.2.8			X	X					X	X			
Shock	4.8.2.4									X	X			
Warm-up time	4.8.2.10					X	X							
Anisoelasticity	4.8.2.9							X	X					
System tests						X	X					X	X	
Flight tests						X	X					X	X	

SCHEDULE 1

QUALIFICATION TEST PLAN

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TEST DESCRIPTION	REFERENCE PARAGRAPH	PRODUCTION SAMPLE			
		1	2	3	
Functional test	4.8.1.2, 4.8.1.3, 4.8.1.4, 4.8.1.5, 4.8.1.6, 4.8.1.7, 4.8.1.8, 4.8.1.9.1	X	X	X	
Multiple cold soak	4.8.2.8		X	X	
Temperature and altitude	4.8.2.6	X			
Vibration endurance	4.8.2.3.2	X			
Shock	4.8.2.4	X			
Anisoelasticity	4.8.2.9	X			
Life test	4.8.2.11		X	X	

SCHEDULE 2

Initial Production Test Plan

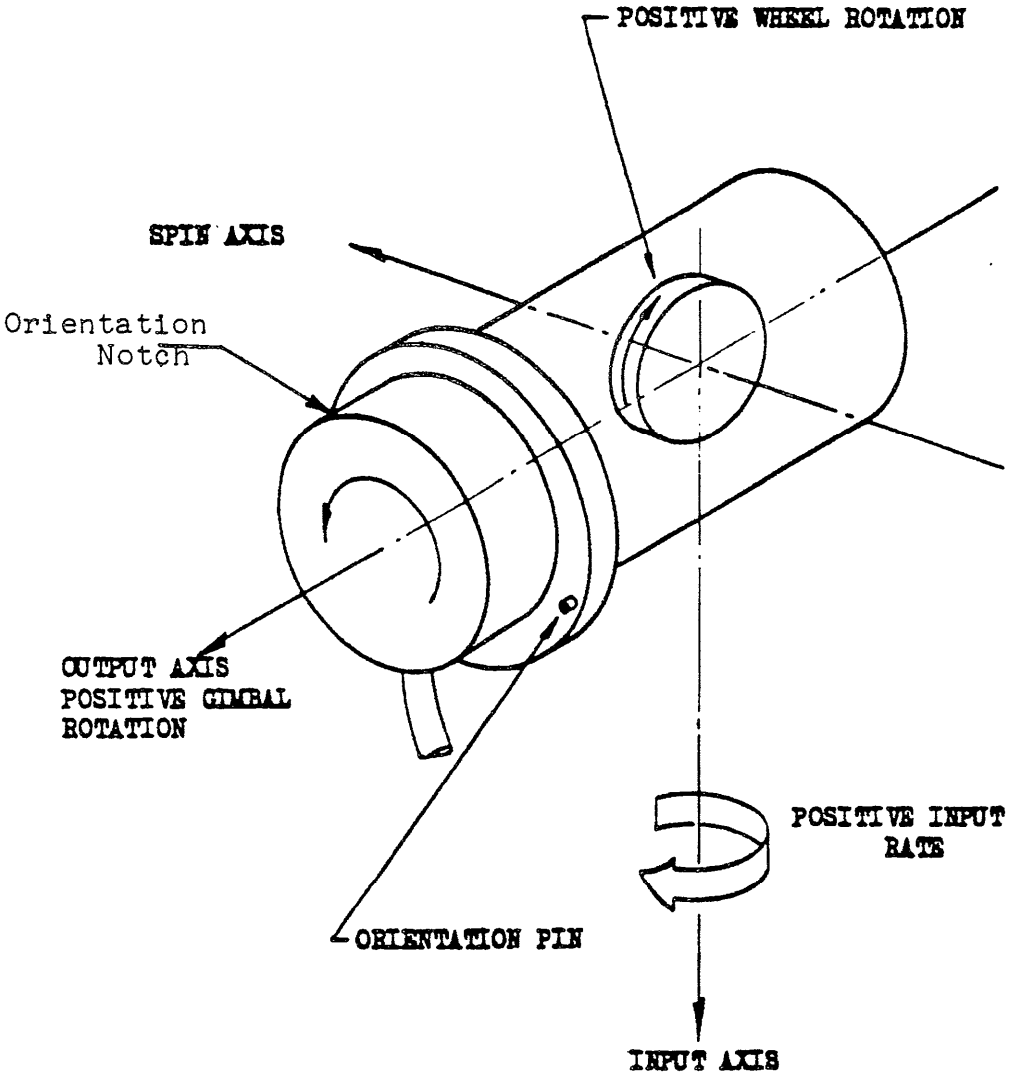
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Check the source to verify that this is the current version before use.

TEST DESCRIPTION	REFERENCE PARAGRAPH	SAMPLE	
		1	
Functional test	4.8.1.2, 4.8.1.3, 4.8.1.4, 4.8.1.5, 4.8.1.6, 4.8.1.7, 4.8.1.8, 4.8.1.9.1	X	
Low temperature operation	4.8.2.1	X	
High temperature operation	4.8.2.2	X	
Vibration sampling	4.8.2.3.1	X	
Shock	4.8.2.4	X	

SCHEDULE 3

PRODUCTION SAMPLING TEST PLAN

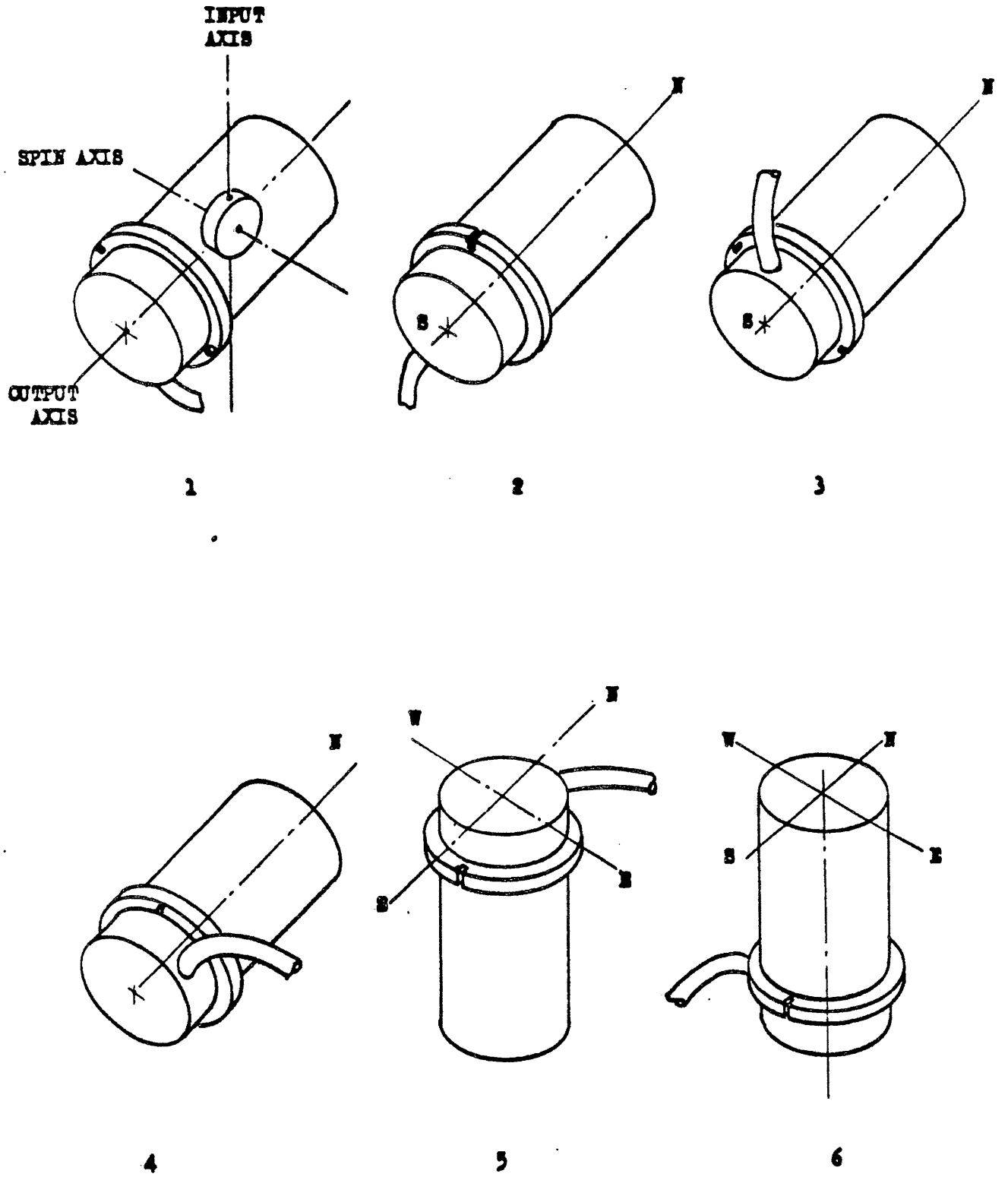
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PHASING DIAGRAM

FIGURE 1

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SIX POSITIONS OF GYRO

FIGURE 2

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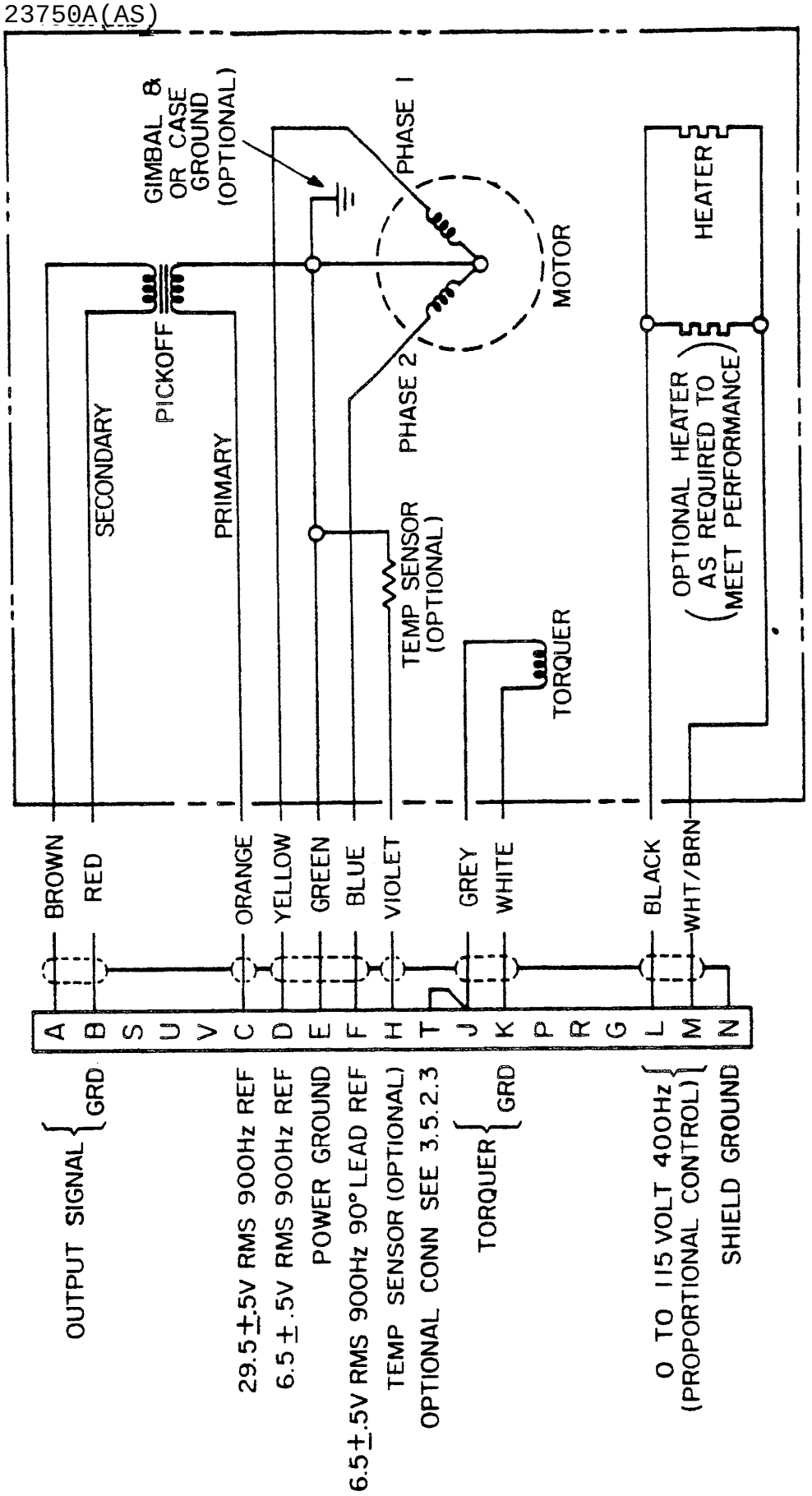
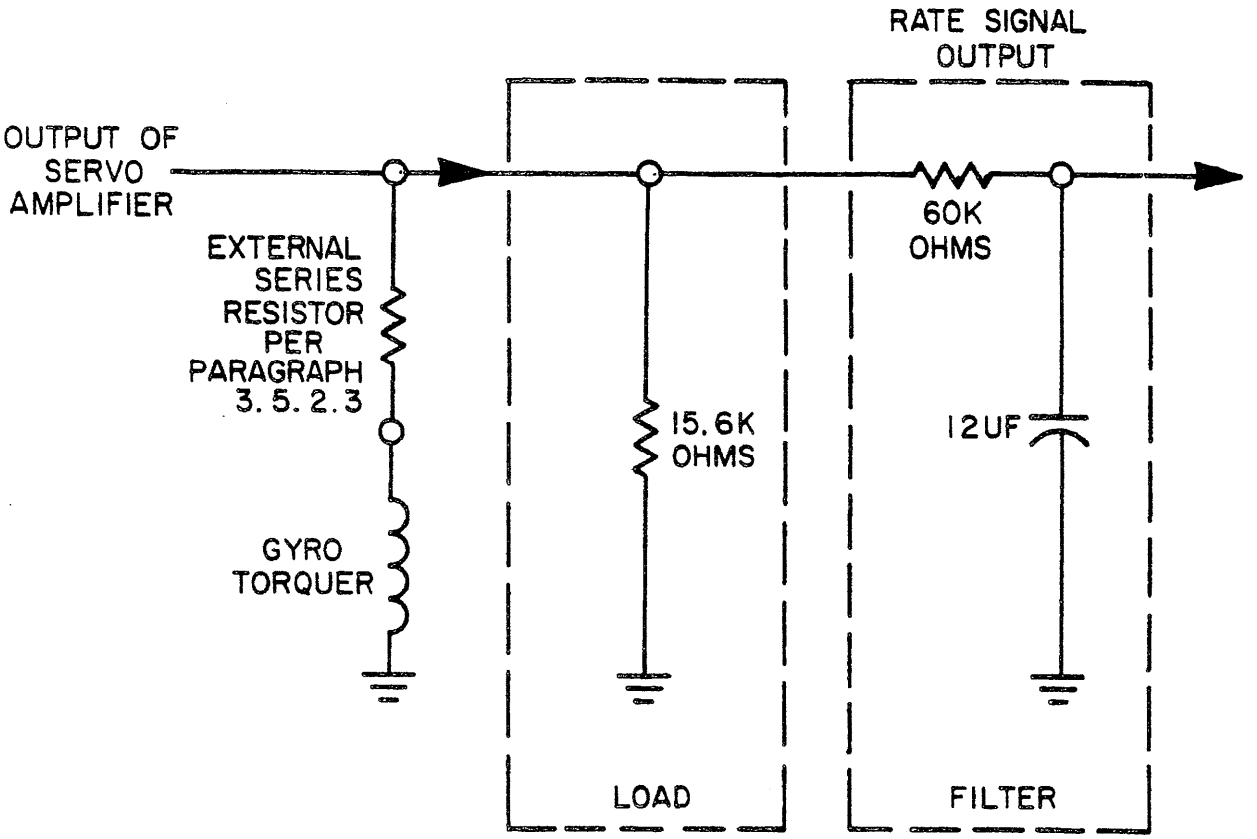


FIGURE 3
ELECTRICAL SCHEMATIC

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NOTE: ALL RESISTORS SHALL BE ± 1 PERCENT TOLERANCE,
ALL CAPACITORS ± 5 PERCENT EXCEPT AS NOTED.

FIGURE 4
OUTPUT LOADS AND FILTER

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SPIN MOTOR

IMPEDANCE LIMITS

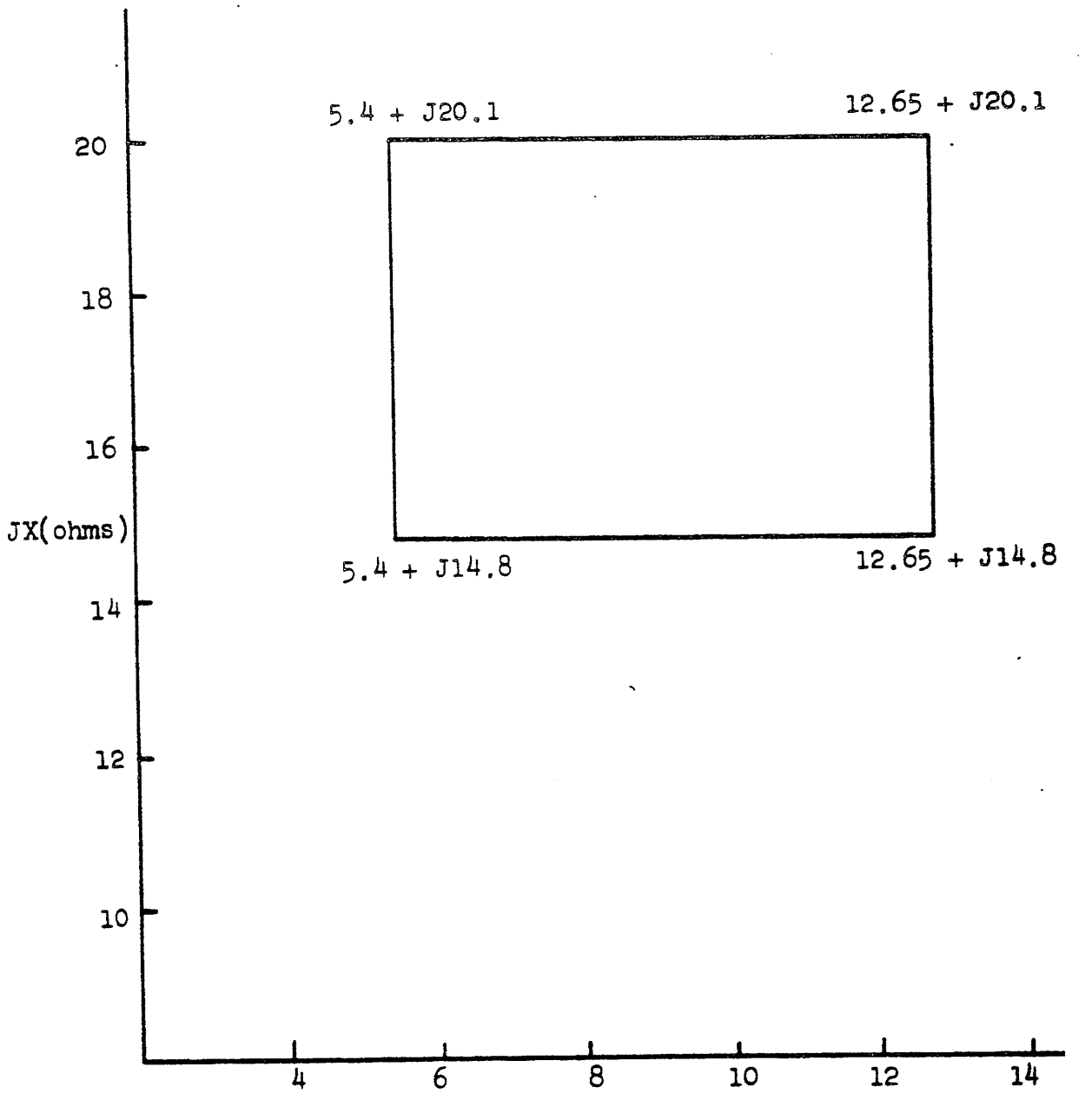


FIGURE 5

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PICKOFF

IMPEDANCE LIMITS

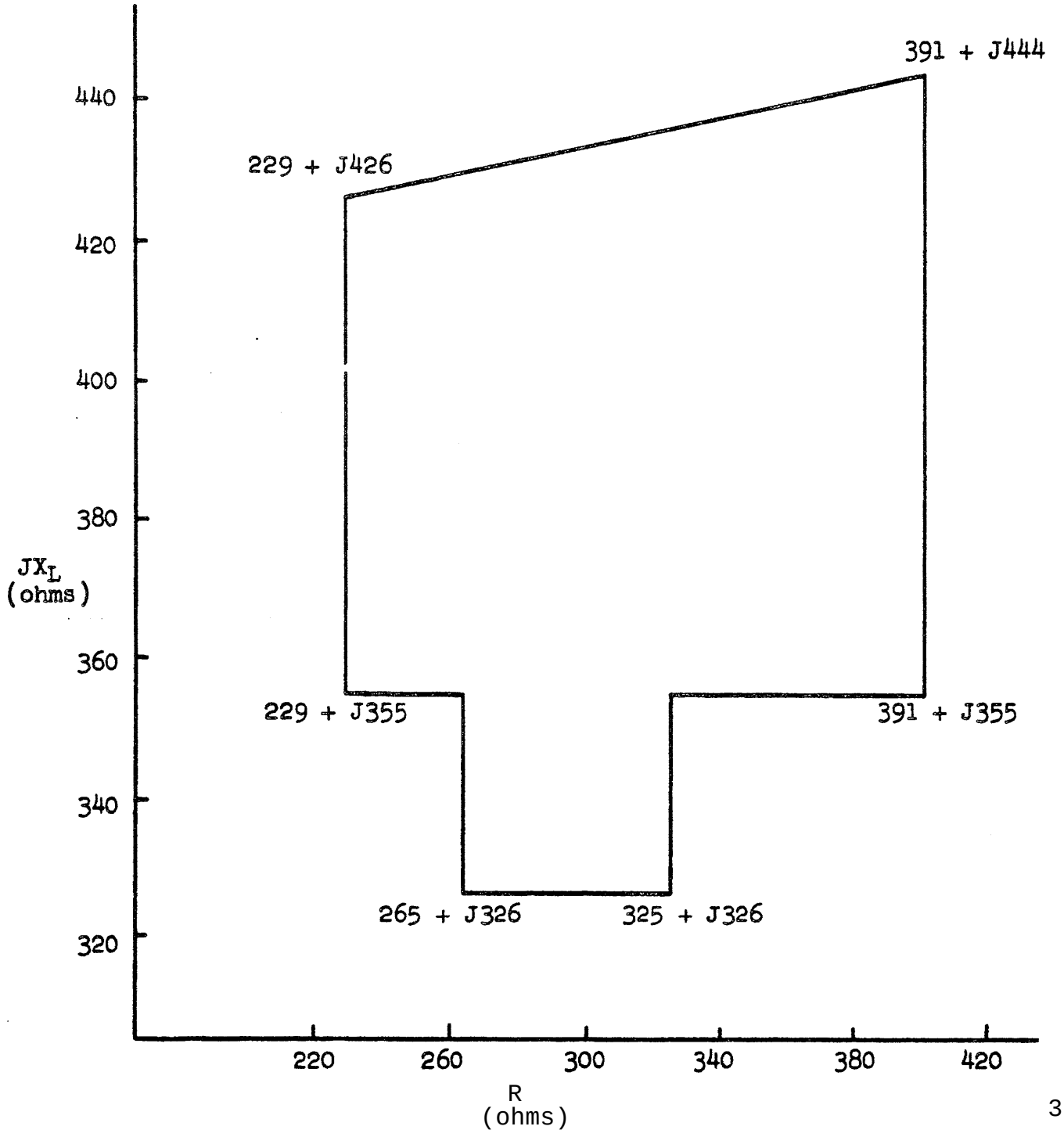


FIGURE 6

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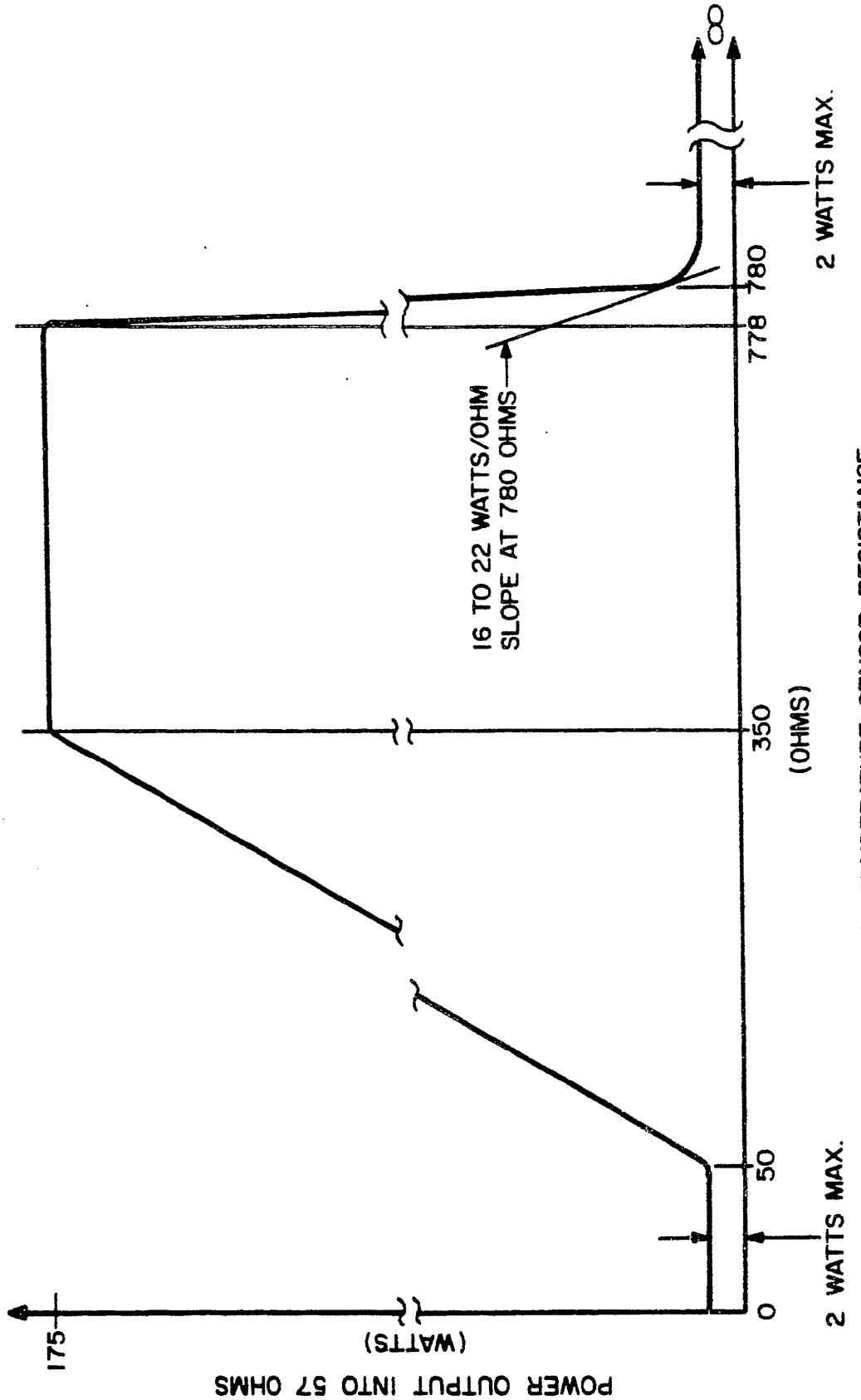


FIGURE 7
HEATER POWER SUPPLY OUTPUT VS TEMPERATURE SENSOR RESISTANCE

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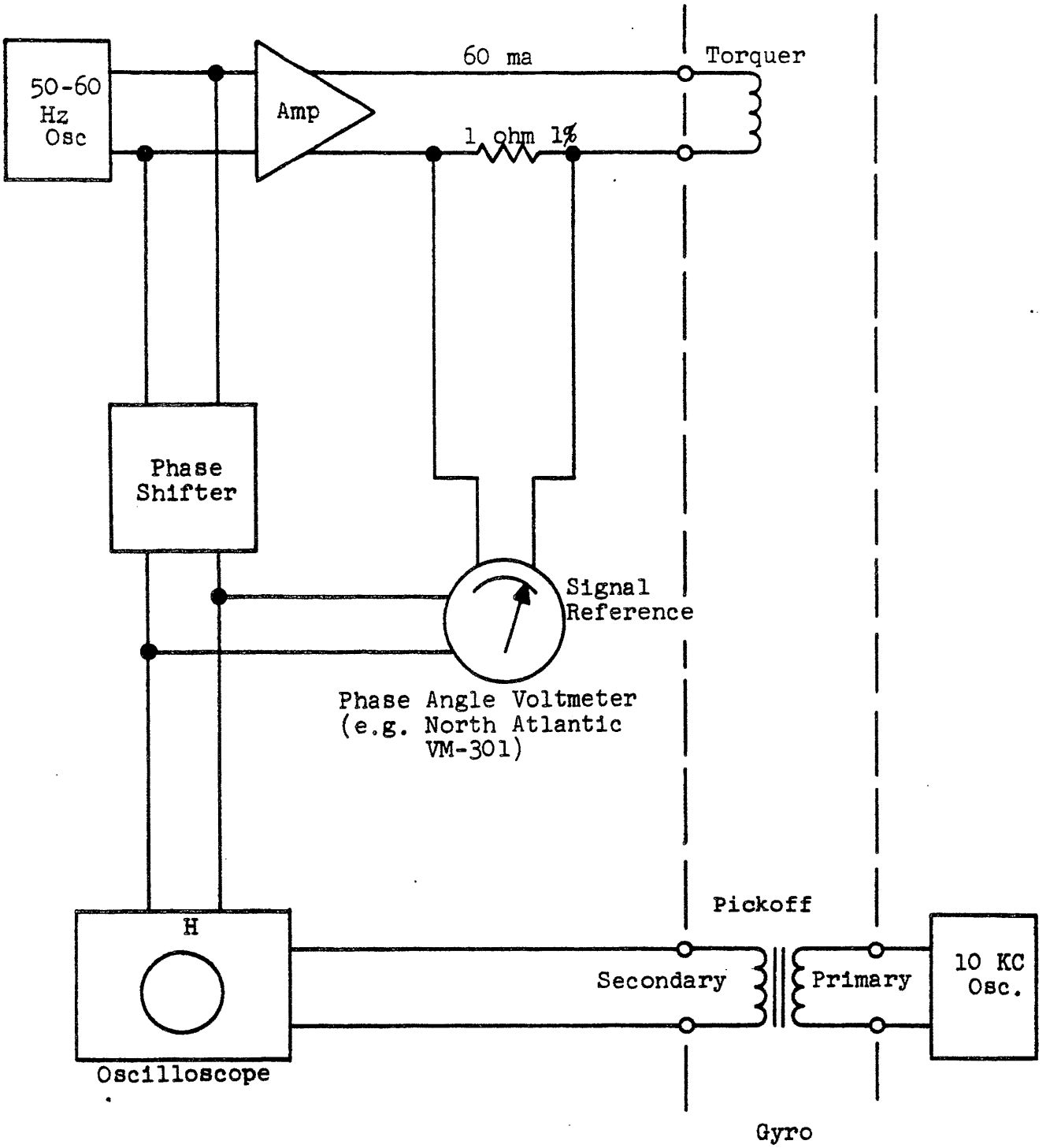


Figure 8 Project Number 1420-N010

Gimbal Corner Frequency Test Circuit

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